Traffic Rule Summary

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1 Introduction

This document serves as an overview about formalized traffic rules based on the German Road Traffic Regulation (StVO), the Vienna Convention on Road Traffic [1] (VCoRT), judicial decisions, and comments by lawyers.

The formalization process of traffic rules in temporal logic can be separated into four steps:

1. **Extracting rules from legal sources**: Based on a traffic rule, all related rules can be extracted from legal sources. Legal sources can contain judicial decisions, other traffic law sections, and consultancy of layers.
2. **Concretizing extracted rules**: The legal texts and judicial decisions are combined and concretized using natural language. As part of the concretization, the situation when the rule is applicable should be expressed. It can also be the case that a single traffic rule addresses different aspects and therefore each has to be considered separately, e.g., StVO $\S 4(1)$ addresses the distance between vehicles and also restricts the braking of a vehicle. The use of natural language allows lawyers to evaluate the concretization so that consistency between the formalized rules for autonomous vehicles and the traffic rules for human traffic participants can be ensured.
3. **Extracting functions, predicates, and propositions**: We use predicates, functions, and atomic propositions for the evaluation of traffic situations, e.g., the allowed speed limit or the velocity of a preceding vehicle. These elements should be defined in a modular way such that they can be easily reused.
4. **Creating temporal logic formulas**: The extracted predicates, functions, and propositions have to be combined to a temporal logic formula matching the concretized sentence from 2).

The remainder of this paper is structured as follows. Sec. 2 introduces mathematical foundations. Sec. **??** presents the formalization of traffic rules. In Sec. **??** predicates and auxiliary functions for the temporal logic formulas are defined.

2 Preliminaries

2.1 Legal information

Please note that no official translated version of the StVO exist. The translation used for this paper is based on the *German Law Archive*[1]. The formalized rules in this work consider the ego vehicle as the autonomous vehicle from whose perception the rules are written. All other traffic participants can either be controlled by humans or can also be autonomous vehicles.

We assume that the considered interstates have no intersections, driving directions are structurally separated so that only lanes with the same driving direction are adjacent, and have right-hand traffic. However, our formalization can also be used for left-hand traffic. Please note that it is possible on German interstates that no speed limit exist. In the following sections, we differentiate main carriage ways, access ramps, exit ramps, and shoulder lanes as potential lane types and solid, dashed, broad solid, and broad dashed lines as possible lane markings.

2.2 Road Network

We describe our road network by lanelets, which are atomic, drivable road segments, geometrically represented by a left and right bound, where the bounds are represented by polylines [2]. The set of all lanelets in a road network is denoted by $\mathbb{L} \cup \{\bot\}$, where $\bot$ is the bottom element for the case that no lanelet exist. Subsequently, we refer to a single lanelet as l, where $l \in \mathbb{L}$. A lanelet can be described by attributes, where the set of all attributes is denoted by $\mathbb{A}$. For interstates, we have $\mathbb{A} = \{access\_ramp, exit\_ramp, main\_carriage\_way, shoulder\}$. The function $\mathrm{attr}(l)$ returns the attribute which is valid for a lanelet (cf. Fig. 1). Similar, the left and right bounds of a lanelet can have lanelet markings. We denote the set of possible lanelet markings as $\mathbb{LM} = \{solid, dashed, broad\_solid, broad\_dashed\}$. The functions $\mathrm{mark_l}(l)$ and $\mathrm{mark_r}(l)$ return the left and right lanelet marking of a lanelet. The functions $\mathrm{l_{lb}}(l)$, $\mathrm{l_{rb}}(l)$, $\mathrm{l_{ini}}(l)$, and $\mathrm{l_{fin}}(l)$ return the left bound, right bound, left and right initial points, and left and right final points of a lanelet. The functions

$$\mathrm{adj_l}(l) = \begin{cases} l' & \text{if } l' \in \mathbb{L} \wedge \mathrm{l_{lb}}(l) = \mathrm{l_{rb}}(l') \\ \bot & \text{otherwise} \end{cases} \tag{1}$$

$$\mathrm{adj_r}(l) = \begin{cases} l' & \text{if } l' \in \mathbb{L} \wedge \mathrm{l_{rb}}(l) = \mathrm{l_{lb}}(l') \\ \bot & \text{otherwise} \end{cases} \tag{2}$$

$$\mathrm{succ}(l) = \begin{cases} l' & \text{if } l' \in \mathbb{L} \wedge \mathrm{l_{fin}}(l) = \mathrm{l_{ini}}(l') \\ & \wedge\ \mathrm{attr}(l) = \mathrm{attr}(l') \\ \bot & \text{otherwise} \end{cases} \tag{3}$$

$$\mathrm{pre}(l) = \begin{cases} l' & \text{if } l' \in \mathbb{L} \wedge \mathrm{l_{ini}}(l) = \mathrm{l_{fin}}(l') \\ & \wedge\ \mathrm{attr}(l) = \mathrm{attr}(l') \\ \bot & \text{otherwise} \end{cases} \tag{4}$$

return the adjacent left, adjacent right, successor, and predecessor lanelet of l, respectively (cf. Fig. 1). The functions $\mathrm{speed\_limit}(ls)$ and $\mathrm{required\_speed}(ls)$, where $ls \subseteq \mathbb{L}$, return a set of maximum and minimum speed limits which a vehicle is allowed to drive on a provided set of lanelets. The functions $\mathrm{ori_l}(l, s_l)$ and $\mathrm{width_l}(l, s_l)$, where s_l is a longitudinal position, return the orientation of the reference path θ_Γ and the width of a lanelet at s_l, respectively.

[1] https://germanlawarchive.iuscomp.org/?p=1290

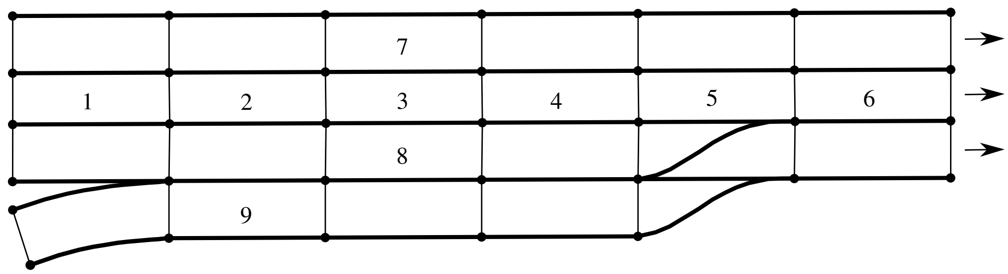

Figure 1: Road network defined by lanelets. The lanelets 1-6 form a lane. Lanelets 2, 4, 7, and 8 are the predecessor, successor, left adjacent, and right adjacent lanelet of lanelet 3, respectively. Lanelets 1-8 are of type $main\_carriage\_way$ and lanelet 9 is of type $access\_ramp$.

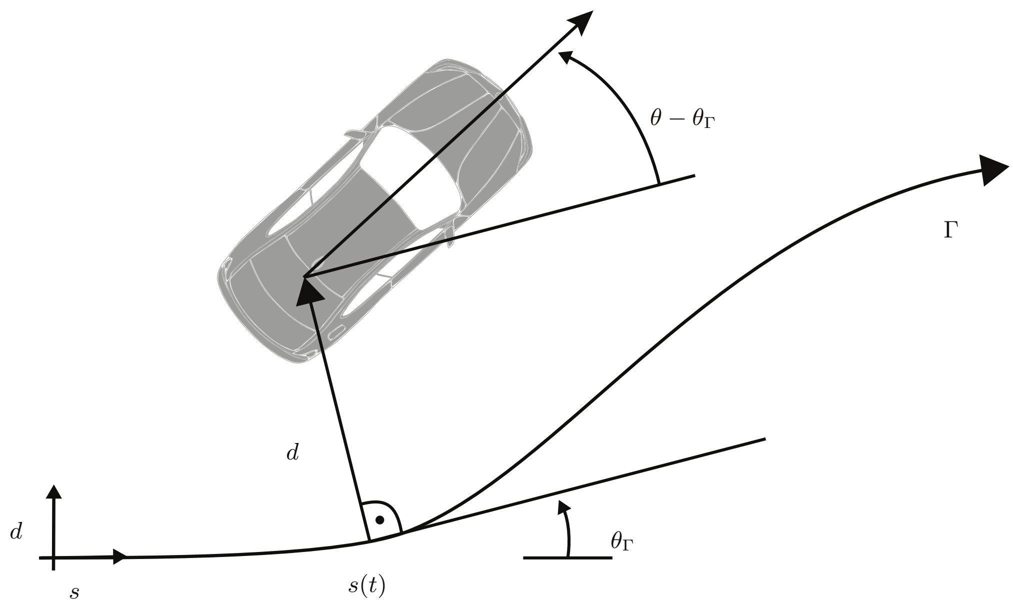

Figure 2: A curvilinear coordinate system aligned to the reference path Γ with orientation θ_Γ. The vehicle is described by the longitudinal position s, the lateral deviation of the reference path d, and the orientation θ.

Definition 1 (Lane)
The lane originating from a lanelet is defined recursively:

$$\begin{aligned}\mathrm{lane}(l) = &\big(\{\mathrm{succ}(l), l, \mathrm{pre}(l)\} \\ &\cup \mathrm{lane}(\mathrm{pre}(l)) \cup \mathrm{lane}(\mathrm{succ}(l))\big) \setminus \{\perp\}\end{aligned}$$

We use a curvilinear curvilinear coordinate system [3] as shown in Fig. 2.

2.3 Vehicle configuration

The state x of a vehicle consists of a longitudinal state $x_{\mathrm{lon}} \in \mathbb{R}^n$ and a lateral state $x_{\mathrm{lat}} \in \mathbb{R}^n$. The longitudinal state $x_{\mathrm{lon}} = \begin{bmatrix} s & v & a & j \end{bmatrix}^T$ consists of position s, velocity v, acceleration a, and jerk j. The lateral state $x_{\mathrm{lat}} = \begin{bmatrix} d & \theta \end{bmatrix}^T$ consists of the lateral distance to the reference path Γ and orientation θ (cf. Fig. 2). The input of the vehicle is denoted by $u(t)$. The underlying vehicle model can be arbitrary as long as a transformation to our state representation is possible. The occupancy of a vehicle for a given state is denoted by $\mathcal{O}(x(t))$. The functions $\mathrm{front}(x(t))$, $\mathrm{rear}(x(t))$, $\mathrm{right}(x(t))$, and $\mathrm{left}(x(t))$ return the position of the front bumper, rear bumper, rightmost point and leftmost point of a vehicle and $v_{\mathrm{type}}(x(t))$ returns the maximum allowed velocity for a vehicle type, e.g., a truck. All variables associated to the ego vehicle are denoted by the subscript $\square_{\mathrm{ego}}$, all variables associated to other vehicles are denote by the subscript $\square_{\mathrm{o}}$, and arbitrary vehicles including the ego vehicle are denoted by the subscript $\square_{\mathrm{k}}$ and $\square_{\mathrm{p}}$, where $\mathrm{o} \in \{1...N\}$, N is the number of other traffic participants, and $\mathrm{k}, \mathrm{p} \in \{0...(N+1)\}$ represent all vehicles in the road network including the ego vehicle. For a given initial state x_0, an input trajectory $u(\cdot)$, we introduce the solution of the model $\dot{x} = f(x, u)$ of the ego vehicle over time t as $\xi(t; x_0, u(\cdot))$. We denote a braking input by $u_{\mathrm{brake}}(\cdot)$. The time at which a safe state is reached for $u(\cdot)$, given the system dynamics and x_0, is denoted by $t_s(u(\cdot))$. We consider a state to be safe on interstates for the ego vehicle if the ego vehicle is in standstill.

2.4 Finite Linear Temporal Logic

We use Finite Linear Temporal Logic (FLTL) since it is interpreted over finite traces of Boolean elements. Given is a set $\mathbb{AP}$ of atomic propositions, where each atomic proposition $\sigma_i \in \mathbb{AP}$ represents a Boolean statement. A FLTL formula ϕ is defined as

$$\begin{aligned} \phi ::= & \, \sigma_i \,|\, \neg\phi \,|\, \phi_1 \wedge \phi_2 \,|\, \phi_1 \vee \phi_2 \,| \\ & X(\phi) \,|\, \overline{X}(\phi) \,|\, \phi_1 U \phi_2 | \, G(\phi) | \, F(\phi) \end{aligned}$$

where the letter X stands for the weak next operator, $\overline{\mathrm{X}}$ for the strong next, U for the until, G for the globally, and F for the future operator. We also require the Boolean operators $\neg$, $\wedge$, and $\vee$. Unary connectives have precedence over binary connectives, $\neg$, X, and $\overline{X}$ have equal precedence and U has precedence over $\wedge$ and $\vee$. The implication $a \implies b$ is defined as $\neg a \vee b$.

We describe the semantics of FLTL informally. For a detailed description of the semantics we refer the reader to [**?**] and [**?**]. The weak next operator expresses that if a next state exists then this state has to satisfy ϕ, whereas the strong next operator expresses that the next state has to exist and that this state has to satisfy ϕ. The until operator says that ϕ_1 is true at least until ϕ_2 is true. The globally operator states that ϕ holds at all states and the future operator states that ϕ holds at some future state.

References

[1] Economic Comission for Europe: Inland Transport Committee, "Vienna Convention on Road Traffic," Nov. 1968.

[2] P. Bender, J. Ziegler, and C. Stiller, "Lanelets: Efficient map representation for autonomous driving," in *Proc. of the IEEE Intelligent Vehicles Symposium*, 2014, pp. 420–425.

[3] E. Héry, S. Masi, P. Xu, and P. Bonnifait, "Map-based curvilinear coordinates for autonomous vehicles," in *Proc. of the IEEE Int. Conf. on Intelligent Transportation Systems*, vol. 2018-March, 2018, pp. 1–7.